Data Analysis and Data Preparation Techniques

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MSCS 634-B01: Advanced Big Data and Data Mining

October 25, 2026

Introduction

Before data can be mined, its basic characteristics must be understood and it must be prepared so that later algorithms produce reliable results. This assignment applies four core preprocessing techniques: descriptive statistics to describe a distribution, distance metrics to measure similarity between objects, smoothing and the interquartile range (IQR) rule to reduce noise and flag outliers, and normalization to place values on comparable scales.

1. Descriptive Statistics and Data Distribution Analysis

Dataset (n = 27, already sorted): 12, 14, 14, 17, 18, 19, 20, 21, 21, 23, 24, 24, 26, 28, 30, 32, 33, 34, 36, 38, 40, 42, 44, 45, 50, 55, 60

a) Mean and median

Sum of all values = 820.

$$\text{Mean} = \Sigma x_i / n = 820 / 27 \approx 30.37$$

With n = 27 (odd), the median is the value at position (n + 1) / 2 = 14. The 14th value is **28**.

Interpretation: the mean (30.37) is larger than the median (28). A few high ages (50, 55, 60) pull the mean upward, so the distribution is positively (right) skewed. Half of the participants are 28 or younger.

b) Mode and modality

Frequency count shows three values occurring twice and every other value occurs once:

Value	Frequency
14	2
21	2
24	2
All others	1

The modes are **14, 21 and 24**. The datasets with one, two, or three modes unimodal, bimodal, and trimodal, so this data is trimodal. Because each mode appears only twice, the modes are weak and do not indicate strong clustering as the ages are spread fairly evenly.

c) Midrange

$$\text{Midrange} = (\min + \max) / 2 = (12 + 60) / 2 = 36$$

The midrange (36) is the midpoint between the youngest and oldest participant. It depends only on the two extremes, so it is sensitive to outliers. Here it exceeds both the mean and the median, again reflecting the long right tail created by the 60-year-old participant.

d) First and third quartiles

Method: split the data at the median (position 14) and take the median of each half, excluding the overall median.

- Lower half (positions 1–13): 12, 14, 14, 17, 18, 19, **20**, 21, 21, 23, 24, 24, 26. Its median is the 7th value, so **Q1 = 20**.
- Upper half (positions 15–27): 30, 32, 33, 34, 36, 38, **40**, 42, 44, 45, 50, 55, 60. Its median is the 7th value, so **Q3 = 40**.

IQR = Q3 − Q1 = 40 − 20 = 20, meaning the middle 50% of participants are between 20 and 40 years old.

e) Five-number summary

Minimum	Q1	Median	Q3	Maximum
12	20	28	40	60

The gap from the median to Q3 (12) is larger than from Q1 to the median (8), and the upper whisker (40 to 60 = 20) is longer than the lower one (12 to 20 = 8). Both confirm the right skew seen in part (a).

2. Distance Metric Calculations

Objects: x = (18, 3, 40, 12) and y = (25, 2, 38, 15). All four metrics start from the attribute-wise absolute differences:

Attribute	x	y	$\|x - y\|$	$\|x - y\|^2$	$\|x - y\|^4$
1	18	25	7	49	2401
2	3	2	1	1	1
3	40	38	2	4	16
4	12	15	3	9	81
Sum			13	63	2499

a) Euclidean distance (L2)

$$d(x, y) = \sqrt{[\ \Sigma\ (x_i - y_i)^2\ ]} = \sqrt{(49 + 1 + 4 + 9)} = \sqrt{63} \approx 7.937$$

This is the straight-line distance between the two points in 4-dimensional space. Squaring the differences gives extra weight to the largest gap (7 on attribute 1).

b) Manhattan distance (L1)

$$d(x, y) = \Sigma\ |x_i - y_i| = 7 + 1 + 2 + 3 = 13$$

Manhattan distance adds the absolute differences, like walking a city grid where you can only move along the axes. It differs from Euclidean distance in three ways:

- It is always ≥ the Euclidean distance (13 vs 7.94), because it cannot take the diagonal shortcut.
- Every unit of difference counts equally. There is no squaring, so one large difference does not dominate.

- It is therefore less sensitive to outliers on a single attribute and often works better in high-dimensional data.

c) Minkowski distance with q = 4 (L4)

$$d(x, y) = (\Sigma |x_i - y_i|^4)^{(1/4)} = (2401 + 1 + 16 + 81)^{(1/4)} = 2499^{(1/4)} \approx 7.071$$

Minkowski distance generalizes the other metrics: q = 1 gives Manhattan, q = 2 gives Euclidean, and $q \to \infty$ gives Supremum. As q increases, the largest difference dominates the sum more and more, so the distance shrinks toward the single biggest gap. Here the results fall steadily as q rises:

q	Metric	Distance
1	Manhattan	13.000
2	Euclidean	7.937
4	Minkowski	7.071
∞	Supremum	7.000

At q = 4 the term from attribute 1 (2401) already makes up about 96% of the sum, so the result (7.071) is nearly the supremum value of 7.

d) Supremum (Chebyshev, L∞) distance

$$d(x, y) = \max_i |x_i - y_i| = \max(7, 1, 2, 3) = 7$$

The supremum distance is the single largest attribute difference it comes from attribute 1. It is most useful when the worst-case deviation on any one attribute matters more than the total difference, for example:

- Quality control or tolerance checks, where a product fails if any one measurement is out of range.
- Anomaly detection, where one extreme attribute should flag a record.
- Grid or board movement where diagonal steps cost the same as straight steps (such as a king in chess or some warehouse robots).

3. Data Smoothing and Outlier Detection

Dataset (n = 28, sorted): 12, 14, 15, 18, 19, 20, 22, 22, 23, 24, 26, 27, 27, 28, 30, 31, 33, 35, 36, 36, 37, 38, 40, 42, 44, 50, 53, 75

a) Smoothing by bin means (bin depth = 4)

Equal-frequency partitioning: 28 values / 4 per bin = 7 bins. Each value is replaced by its bin's mean.

Bin	Original values	Sum	Mean (Sum / 4)	Smoothed values
1	12, 14, 15, 18	59	14.75	14.75, 14.75, 14.75, 14.75
2	19, 20, 22, 22	83	20.75	20.75, 20.75, 20.75, 20.75
3	23, 24, 26, 27	100	25.00	25, 25, 25, 25
4	27, 28, 30, 31	116	29.00	29, 29, 29, 29
5	33, 35, 36, 36	140	35.00	35, 35, 35, 35
6	37, 38, 40, 42	157	39.25	39.25, 39.25, 39.25, 39.25
7	44, 50, 53, 75	222	55.50	55.5, 55.5, 55.5, 55.5

Effect on the data and observations:

- Local variation inside each bin disappears. The data now has only 7 distinct values, which reduces noise and simplifies later analysis.
- The overall mean is unchanged (877 / 28 ≈ 31.32 before and after), because each bin's total is preserved.
- The bin means rise steadily (14.75 → 55.5), so the increasing trend of the data is kept.
- Steps between bins are fairly even (about 4–6) until bin 7, which jumps by 16.25. The outlier 75 inflates that bin's mean: without it, the other three values average 49. Binning by means is therefore sensitive to outliers, which bleed into their neighbors.

b) Outlier detection with the IQR method

With n = 28 (even), the median is the average of the 14th and 15th values: (28 + 30) / 2 = 29. The data splits into two halves of 14 values each.

- Lower half: 12, 14, 15, 18, 19, 20, **22, 22**, 23, 24, 26, 27, 27, 28. Q1 = (22 + 22) / 2 = **22**.
- Upper half: 30, 31, 33, 35, 36, 36, **37, 38**, 40, 42, 44, 50, 53, 75. Q3 = (37 + 38) / 2 = **37.5**.

$$\text{IQR} = Q3 - Q1 = 37.5 - 22 = 15.5$$

$$1.5 \times \text{IQR} = 23.25$$

$$\text{Lower fence} = Q1 - 1.5 \times \text{IQR} = 22 - 23.25 = -1.25$$

$$\text{Upper fence} = Q3 + 1.5 \times \text{IQR} = 37.5 + 23.25 = 60.75$$

Following the common rule of thumb, any value more than 1.5 × IQR below Q1 or above Q3 is a potential outlier, so the cutoffs are values below −1.25 or above 60.75.

Result: No value is below −1.25. Only **75** exceeds 60.75, so 75 is the single outlier. 53 is the largest non-outlier.

Significance: 75 lies 22 units beyond the next-highest value (53). It may be a data-entry error, a measurement problem, or a genuine but rare case. It should be investigated before it is removed, capped, or kept. Left untreated, it raises the mean (31.32 vs 29.70 without it) and, as seen in part (a), distorts the last bin mean.

c) Alternative smoothing techniques

1. Smoothing by bin boundaries. Sort and partition the data into bins as before. Find each bin's minimum and maximum, then replace every value with whichever boundary is closer. For example, bin 2 (19, 20, 22, 22) becomes 19, 19, 22, 22. It is preferable to bin means when the smoothed values must remain real, observed values (for example, discrete counts or category codes) and when sharper bin edges matter more than an average. Bin medians are a related variant that resists outliers better than bin means.

2. Regression smoothing. Fit a function, such as a linear or polynomial regression line, that captures the relationship between the attribute and another variable (or its position/time). Each noisy value is replaced by the fitted value on the curve. It is preferable when the data follows a clear trend or depends on another attribute, such as sales over time or temperature versus energy use. It keeps the overall pattern instead of creating flat steps the way binning does.

Other options include moving averages for time series and clustering, where values are smoothed toward cluster centers and points far from any cluster are treated as outliers.

4. Data Normalization Techniques and Calculations

Dataset: 150, 250, 350, 500, 800. Base statistics used below:

- n = 5, minimum = 150, maximum = 800, range = 650
- Mean: (150 + 250 + 350 + 500 + 800) / 5 = 2050 / 5 = **410**
- Deviations from the mean: −260, −160, −60, 90, 390

a) Min-max normalization to [0, 1]

$$v' = (v - \min) / (\max - \min) \times (\text{new_max} - \text{new_min}) + \text{new_min} = (v - 150) / 650$$

v	**v − 150**	**v′**
150	0	0.0000
250	100	0.1538
350	200	0.3077
500	350	0.5385
800	650	1.0000

Min-max normalization is a linear rescale that keeps the relative spacing between values. The minimum maps to 0 and the maximum to 1. It is sensitive to outliers, and new data outside the original range would fall outside [0, 1].

b) Z-score normalization with standard deviation

Squared deviations: 67,600 + 25,600 + 3,600 + 8,100 + 152,100 = 257,000.

$$\sigma = \sqrt{[\ \Sigma\ (v - \text{mean})^2 / n\ ]} = \sqrt{(257{,}000 / 5)} = \sqrt{51{,}400} \approx 226.72$$

$$v' = (v - 410) / 226.72$$

v	**v − 410**	**(v − 410)²**	**v′**
150	−260	67,600	−1.1468
250	−160	25,600	−0.7057
350	−60	3,600	−0.2646
500	90	8,100	0.3970
800	390	152,100	1.7202

Each z-score is the number of standard deviations a value lies from the mean. The normalized values have mean 0 and standard deviation 1. This method works when the true min and max are unknown, and it is less distorted by outliers than min-max. (This uses the population standard deviation, dividing by n. With the sample standard deviation, $s = \sqrt{(257{,}000 / 4)} \approx 253.48$, the scores would be −1.0257, −0.6312, −0.2367, 0.3551 and 1.5386.)

c) Z-score normalization with mean absolute deviation

Absolute deviations: 260 + 160 + 60 + 90 + 390 = 960.

$$s_A = (1/n)\ \Sigma\ |v - \text{mean}| = 960 / 5 = 192$$

$$v' = (v - 410) / 192$$

v	**v − 410**	**|v − 410|**	**v′**
150	−260	260	−1.3542
250	−160	160	−0.8333
350	−60	60	−0.3125
500	90	90	0.4688
800	390	390	2.0313

Mean absolute deviation does not square the deviations, so an extreme value such as 800 has less influence on the scale. That makes this variant more robust to outliers than the standard-deviation version. Since 192 < 226.72, the scores spread farther from 0.

d) Normalization by decimal scaling

$v' = v / 10^j$, where j is the smallest integer such that $\max(|v'|) < 1$

The largest absolute value is 800. Dividing by $10^3 = 1000$ gives 0.8 < 1, while 10^2 would give 8, so **j = 3**.

v	**v / 1000**
150	0.15
250	0.25
350	0.35
500	0.50
800	0.80

Decimal scaling moves the decimal point so all values fall within (−1, 1). It is simple and keeps the original proportions, but it does not center the data, and the result depends on the single largest magnitude.

Summary of normalization results

Original	Min-max	Z-score (SD)	Z-score (MAD)	Decimal scaling
150	0.0000	−1.1468	−1.3542	0.15
250	0.1538	−0.7057	−0.8333	0.25
350	0.3077	−0.2646	−0.3125	0.35
500	0.5385	0.3970	0.4688	0.50
800	1.0000	1.7202	2.0313	0.80

Conclusion

The four exercises show how each preprocessing step reveals or corrects something about the data. The age data is right-skewed (mean 30.37 > median 28), and the five-number summary of 12, 20, 28, 40, 60 shows where that skew sits. The distance results (13, 7.94, 7.07, 7) show that a higher Minkowski order puts more weight on the single largest attribute difference. In the smoothing data, the IQR rule isolates 75 as the only outlier, and the same value distorts the last bin mean, which explains why outliers are usually handled before smoothing. Finally, the normalization methods map the same values onto different scales: min-max preserves the original spacing, z-scores center the data, MAD-based z-scores reduce outlier influence, and decimal scaling simply shifts the decimal point. The right choice depends on whether the downstream algorithm is sensitive to scale, centering, or extreme values.

References

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